

The Technology Value Stream in DevOps

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What is a Technology Value Stream?

A Technology Value Stream represents all activities required to deliver technology products and services from concept to customer.

It includes:

- Planning
- Development
- Testing
- Deployment
- Operations
- Customer Feedback

The goal is to improve flow, eliminate waste, and deliver value faster.

Technology Value Stream Process



Benefits:

- Faster delivery
- Improved quality
- Better collaboration
- Reduced waste

Defining Lead Time

Lead Time:

Lead Time is the total amount of time from when work is requested until it is delivered to the customer.

Lead Time includes:

- Waiting time
- Queue time
- Development time
- Testing time
- Deployment time

Formula:

Lead Time = Request to Delivery

Defining Processing Time

Processing Time:

Processing Time is the amount of time spent actively working on a task.

Examples:

- Coding
- Testing
- Reviewing code
- Deploying software

Processing Time Excludes:

- Waiting
- Delays
- Approvals
- Queue time

Lead Time vs. Processing Time

Measurement	Lead Time	Processing Time
Includes Waiting?	Yes	No
Includes Active Work?	Yes	Yes
Customer Focused?	Yes	No
Measures Entire Process?	Yes	No

Example:

- Lead Time = 30 Days
- Processing Time = 8 Hours

Most delays occur while work waits between stages.

Common Scenario – Deployment Lead Times Requiring Months

Traditional organizations often experience deployment lead times measured in months because of:

- Manual processes
- Approval bottlenecks
- Large software releases
- Siloed teams
- Lengthy testing cycles

Consequences:

- Slower innovation
- Increased risk
- Customer dissatisfaction

Our DevOps Ideal – Deployment Lead Times of Minutes

DevOps organizations strive for deployment lead times measured in minutes.

This is achieved through:

- Continuous Integration (CI)
- Continuous Delivery (CD)
- Automation
- Automated Testing
- Infrastructure as Code
- Continuous Monitoring

Benefits:

- Faster releases
- Improved reliability
- Faster customer feedback

Why Reducing Lead Time Matters

Reducing lead time helps organizations:

- Deliver customer value faster
- Improve software quality
- Reduce deployment risk
- Respond quickly to market changes
- Increase customer satisfaction

Companies that deploy frequently often outperform competitors.

Conclusion

Key Takeaways:

- Technology Value Streams focus on delivering value efficiently.
- Lead Time measures total delivery time.
- Processing Time measures active work time.
- Traditional deployments may take months.
- DevOps aims for deployments measured in minutes.
- Automation and continuous delivery are essential DevOps practices.

Thank You!!!

References

- Forsgren, N., Humble, J., & Kim, G. (2018). *Accelerate: The science of lean software and DevOps*. IT Revolution Press.
- Kim, G., Humble, J., Debois, P., & Willis, J. (2021). *The DevOps handbook* (2nd ed.). IT Revolution Press.
- Kersten, M. (2018). *Project to product*. IT Revolution Press.